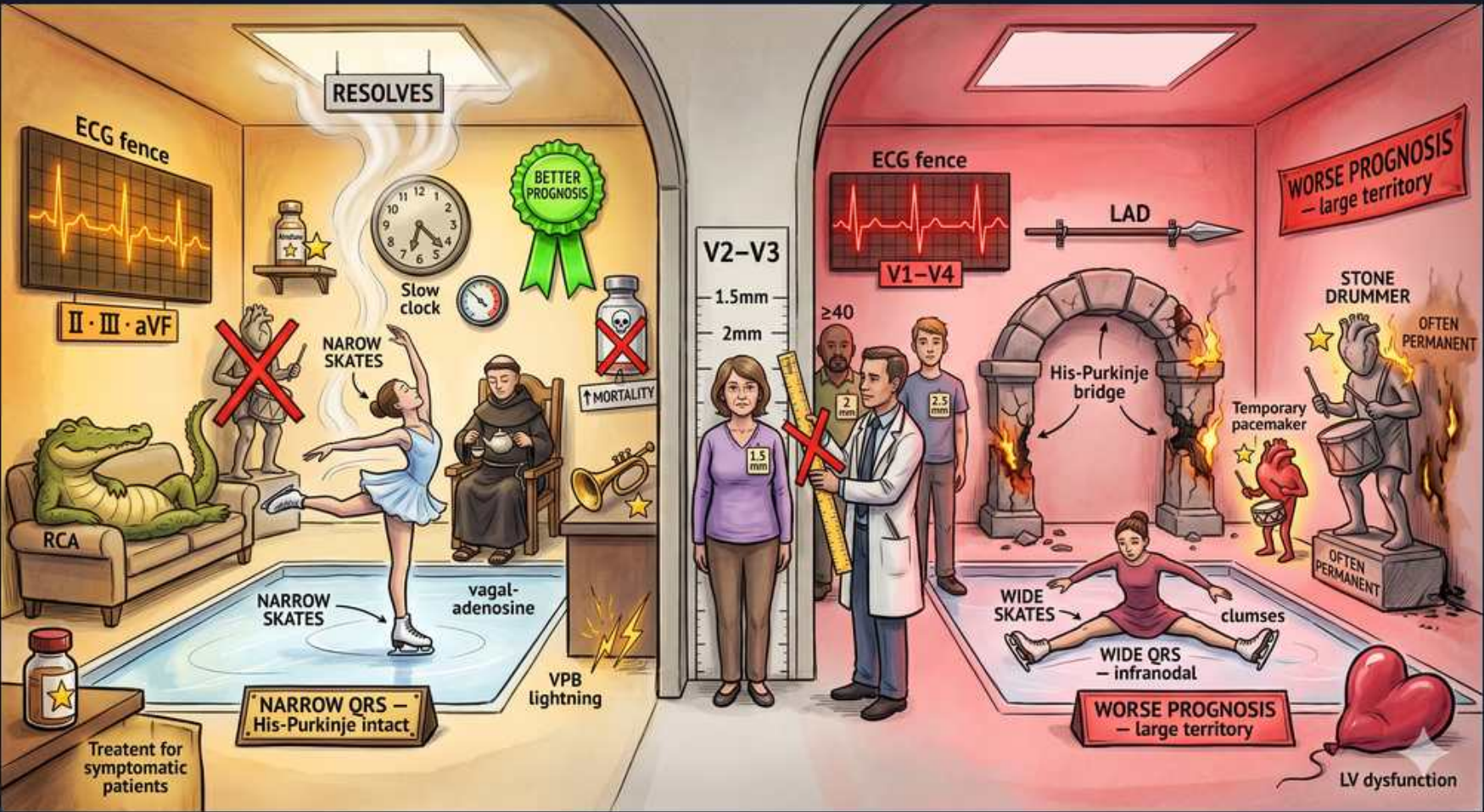


STEMI — Conduction Blocks: Narrow vs Wide QRS & Prognosis



1. Inferior MI — II, III, aVF (RCA)

WHAT YOU SEE

'ECG fence II-III-aVF' + RCA bottle + crocodile

WHAT IT ENCODES

Inferior STEMI = ST elevation in II, III, aVF, usually RCA occlusion. AV-nodal artery off the RCA → AV block. Inferior-MI blocks are typically NARROW-complex, supranodal, transient, and benign.

BOARD TRAP

Inferior MI AV block is usually narrow-complex and transient — don't assume it needs a permanent pacemaker.

2. Narrow QRS — His-Purkinje intact

WHAT YOU SEE

'NARROW QRS — His-Purkinje intact' (figure skater)

WHAT IT ENCODES

Narrow-QRS escape (junctional, ~40–60 bpm) means the block is at the AV node, above the His bundle — the His-Purkinje system is intact, escape is stable and responsive to atropine.

BOARD TRAP

A narrow escape rhythm is a reassuring, AV-nodal (supranodal) block — usually transient with inferior MI.

3. Resolves — slow clock, better prognosis

WHAT YOU SEE

'RESOLVES' steam + 'SLOW CLOCK BETTER PROGNOSIS' + ↓mortality

WHAT IT ENCODES

AV-nodal block from inferior MI typically resolves over hours-days with reperfusion; lower mortality. Often needs only observation ± atropine; temporary pacing if symptomatic bradycardia.

BOARD TRAP

Permanently pacing a self-resolving inferior-MI AV block — most recover; pacing is usually temporary.

4. Vagal / adenosine-like, atropine-responsive

WHAT YOU SEE

'VAGAL-ADENOSINE' monk figure

WHAT IT ENCODES

AV-nodal block is vagally mediated and adenosine-sensitive physiology — it responds to atropine. This contrasts with infranodal block, which is atropine-unresponsive.

BOARD TRAP

Giving atropine for an infranodal (wide-QRS) block — it won't help and may paradoxically worsen the rate.

5. VPB lightning — treat if symptomatic

WHAT YOU SEE

'VPB LIGHTNING — Treat for symptomatic patients'

WHAT IT ENCODES

Ventricular premature beats post-MI are common; treat only if symptomatic/hemodynamically significant. Prophylactic lidocaine for asymptomatic PVCs is NOT indicated (increases mortality).

BOARD TRAP

Prophylactic antiarrhythmics for asymptomatic post-MI PVCs — CAST trial showed increased mortality; treat only symptomatic arrhythmia.

6. Anterior MI — V1–V4 (LAD)

WHAT YOU SEE

'ECG fence V1–V4, LAD' (right red panel)

WHAT IT ENCODES

Anterior STEMI = ST elevation V1–V4, LAD occlusion, large territory. New bundle-branch/bifascicular block here reflects extensive septal infarction with infranodal conduction damage.

BOARD TRAP

A new bifascicular or wide-complex block in anterior MI signals a large infarct — high risk for complete heart block.

7. His-Purkinje bridge / infranodal

WHAT YOU SEE

'His-Purkinje bridge' archway on fire + temporary pacemaker

WHAT IT ENCODES

Anterior MI damages the infranodal His-Purkinje system → wide-QRS block. Escape is unreliable, slow, and atropine-unresponsive → needs temporary (often permanent) pacing.

BOARD TRAP

Treating an infranodal His-Purkinje block with atropine/observation — it needs pacing because the escape is unstable.

8. Wide QRS — infranodal, clumsy escape

WHAT YOU SEE

'WIDE SKATES — infranodal / clumsy' figure

WHAT IT ENCODES

A wide-QRS escape rhythm localizes the block BELOW the His bundle (infranodal). The ventricular escape is slow (~20–40), unreliable, and prone to asystole — pacemaker indicated.

BOARD TRAP

Wide-complex escape = infranodal = dangerous; do not wait for spontaneous recovery as you might with AV-nodal block.

9. Worse prognosis — large territory

WHAT YOU SEE

'WORSE PROGNOSIS — large territory'

WHAT IT ENCODES

Conduction block in anterior MI marks a large infarct → higher mortality, worse LV function, often permanent block. Contrast with the benign, transient inferior-MI nodal block.

BOARD TRAP

Equating anterior-MI block (large territory, often permanent) with the benign transient inferior-MI block.

10. Stone drummer — often permanent

WHAT YOU SEE

'STONE DRUMMER — OFTEN PERMANENT'

WHAT IT ENCODES

Infranodal block from anterior MI frequently does not recover and requires permanent pacemaker implantation, unlike the self-resolving AV-nodal block of inferior MI.

BOARD TRAP

Assuming all post-MI blocks recover — anterior-MI infranodal block is often permanent and needs a permanent device.

11. Temporary pacemaker

WHAT YOU SEE

'Temporary pacemaker' icon

WHAT IT ENCODES

Indications for temporary pacing in acute MI: symptomatic bradycardia unresponsive to atropine, Mobitz II, complete heart block, new bifascicular/alternating BBB — bridge to permanent device.

BOARD TRAP

Withholding temporary pacing in Mobitz II or new bifascicular block during anterior MI — these herald complete block.

12. LV dysfunction

WHAT YOU SEE

Broken heart 'LV DYSFUNCTION'

WHAT IT ENCODES

Large anterior infarcts with conduction disease often leave significant LV dysfunction → consider ICD/CRT evaluation (after 40-day post-MI waiting period if EF ≤35%).

BOARD TRAP

Implanting an ICD immediately post-MI — wait ~40 days and reassess EF before primary-prevention ICD.

13. V2–V3 ST elevation thresholds

WHAT YOU SEE

'V2–V3: 1.5mm / 2mm / ≥40' ruler figures

WHAT IT ENCODES

STEMI ST-elevation thresholds in V2–V3 are age/sex-specific: ≥2.5 mm men <40, ≥2 mm men ≥40, ≥1.5 mm women. Other leads ≥1 mm. Correct thresholds avoid over/under-calling STEMI.

BOARD TRAP

Applying the 1 mm rule to V2–V3 — these leads need higher, sex/age-specific cutoffs (2–2.5 mm men, 1.5 mm women).
